$$N_{push} = mg + F\cos\theta \qquad (3.29)$$

As a result the maximal static friction also increases and is equal to

$$f_s^{max} = \mu_s N_{push} = \mu_s \left(mg + F\cos\theta\right) \qquad (3.30)$$

Equation (3.30) shows that a greater force needs to be applied to push the object into motion.

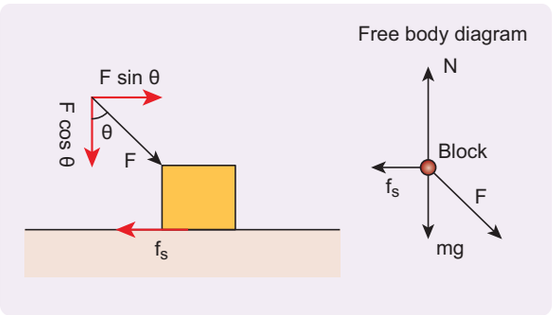


Figure 3.26 An object is pushed at an angle θ

When an object is pulled at an angle θ, the applied force is resolved into two components as shown in Figure 3.27. The total downward force acting on the object is

$$N_{pull} = mg - F\cos\theta \qquad (3.31)$$

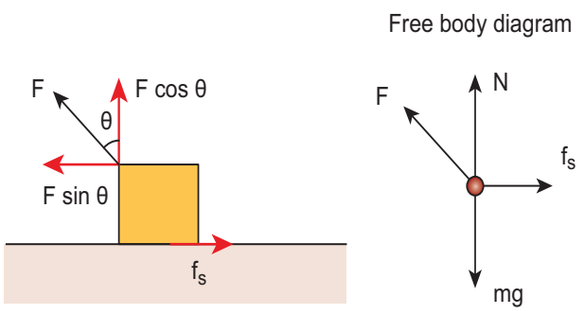


Figure 3.27 An object is pulled at an angle θ

Equation (3.31) shows that the normal force is less than N_{push}. From equations (3.29) and (3.31), it is easier to pull an object than to push to make it move.

3.6.5 Angle of Friction

The angle of friction is defined as the angle between the normal force (N) and the resultant force (R) of normal force and maximum friction force (f_s^{max})

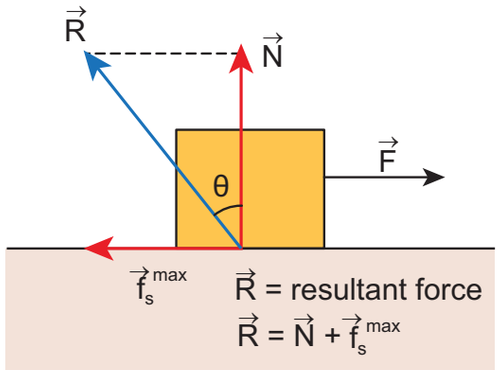


Figure 3.28 Angle of Friction

In Figure 3.28 the resultant force is $R = \sqrt{\left(f_s^{max}\right)^2 + N^2}$

$$\tan\theta = \frac{f_s^{max}}{N} \qquad (3.32)$$

But from the frictional relation, the object begins to slide when $f_s^{max} = \mu_s N$

$$\text{or when } \frac{f_s^{max}}{N} = \mu_s \qquad (3.33)$$

From equations (3.32) and (3.33) the coefficient of static friction is

$$\mu_s = \tan\theta \qquad (3.34)$$

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